

Why Are We Losing Customers?

Churn Diagnosis & Retention Strategy for a B2B Project Management SaaS Company

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Digital Product Analytics Portfolio
SaaS Customer Churn Prediction Project

1. Project Topic & Overview

This project investigates customer churn for a B2B Project Management SaaS company, combining root-cause diagnosis with a data-driven retention strategy. It is built on a synthetic datasets of 10,000 customer records across a twelve-table schema `dim_customers`, `dim_date`, `dim_complaint_theme`, `fact_usage_monthly`, `fact_engagement_monthly`, `fact_customer_monthly_snapshot`, `fact_payment_attempts`, `fact_support_tickets`, `fact_surveys`, `fact_churn_predictions`, `customer_churn_raw`, and `customer_churn_clean` — analyzed using PostgreSQL for data modeling and diagnostic querying.

The analysis is anchored by two headline findings: an overall churn rate of 10.21%, with 28.40% of all churn concentrated within customers' first six months; and near-identical ARPU between churned (\$34.57) and retained (\$34.97) customers. Together these findings re-frame the core question of the project churn in this business is experience-driven rather than price-driven and set the direction for both the diagnostic investigation and the retention recommendations that follow.

2. Key Business Objectives

Four objectives anchor the project, spanning both the diagnostic and strategic halves of the title.

Objective 1: Diagnose the root causes of customer churn

Identify the behavioral, engagement, and support-related factors that most strongly differentiate churned from retained customers moving beyond assumption (e.g., “customers leave because of price”) to evidence, using the ARPU parity finding as proof that churn is experience-driven, not price-driven.

Objective 2: Quantify and prioritize churn risk across the active customer base

Build a systematic way to identify which current customers are most likely to churn and how much revenue is at stake, so retention resources can be targeted at the highest-impact accounts rather than spread evenly across the whole base.

Objective 3: Pinpoint the critical early-tenure window driving disproportionate churn

Given that 28.1% of churn happens in the first six months, isolate what breaks down during on-boarding and early engagement, and define the specific intervention points (tenure milestones, engagement thresholds) where the business can act before a customer disengages for good.

Objective 4: Translate diagnostic and risk insights into an actionable retention strategy

Convert the findings from Objectives 1–3 into concrete, prioritized recommendations — intervention triggers, playbooks by complaint/risk type, and resource allocation guidance — with an estimate of the revenue impact of acting on them.

Objective-to-Analytic Alignment

Each objective is answered by a specific combination of the analytic categories detailed in Section 3.

Objective	Focus	Analytic Drawn On Categories
Objective 1	Diagnose the root causes of customer churn	Descriptive + Diagnostic
Objective 2	Quantify and prioritize churn risk across the active customer base	Risk Analysis + Predictive
Objective 3	Pinpoint the critical early-tenure window driving disproportionate churn	Diagnostic + Risk Analysis
Objective 4	Translate diagnostic and risk insights into an actionable retention strategy	Prescriptive Analytic

3. Analytic Framework

The project follows a structured Digital Product Analytic arc from baseline measurement to diagnosis, risk quantification, predictive evaluation and prescriptive retention action.

- Descriptive Analytics: baseline churn, customer mix, ARPU, engagement, support, payments, tenure and revenue concentration.
- Diagnostic Analytics: behavioral, usage, support, satisfaction, payment and lifecycle comparisons between churned and retained customers.
- Risk Analysis: current high-risk accounts, risk concentration, tenure windows, segment exposure and revenue at risk.
Predictive Analytic: analysis of existing churn-probability outputs; no new model is trained in the supplied SQL.
- Prescriptive Analytic: intervention thresholds, account prioritization, complaint playbooks and retention resource allocation.

4. Product-Driven Business Questions

The business questions below are organized into five analytic categories, forming a structured arc from establishing what is happening, through explaining why, to quantifying exposure, forecasting what's next, and prescribing action.

4.1 Descriptive Analytic — “What is happening?”

- What is the overall churn rate, and how has it trended over time (monthly/quarterly)?
- How is the customer base distributed by customer_segment, gender, city, and country?
- What does average revenue per user (ARPU) look like across segments?
- What are the average engagement scores, support ticket volumes, and payment failure rates across the customer base?

- What is the distribution of customer tenure at the point of churn?
- What proportion of revenue is concentrated in which customer segments?

4.2 Diagnostic Analytic — “Why is it happening?”

- Why do churned and retained customers show near-identical ARPU (~\$34.57 vs \$34.97) what does this rule out about churn being price-driven?
- What factors differentiate churned customers from retained customers, and which of these differences are most significant?"
- Is there a relationship between customer ARPU, pricing, and churn, and can spending level be used to identify customers at higher risk of churn?"
- Why is churn concentrated so heavily (28.40%) in the first six months what on-boarding or early-engagement failures explain this?
- Which support ticket types/complaint themes correlate most strongly with churn?
- Do payment failures precede churn, or are they a symptom of disengagement rather than a cause?
- Is there a relationship between engagement score decline and churn timing?
- How do churned customers' usage patterns differ from retained customers' in the months leading up to churn?
- Are there country-level or city-level differences in ticket volume, complaint type, or churn rate?

4.3 Risk Analysis — “Where is the exposure, and how severe is it?”

- Which active customers currently show the highest composite behavioral risk scores (engagement + tickets + payment failures)?
- What percentage of current MRR sits with “high risk” accounts on the watch-list?
- Which tenure milestones (30/60/90/180 days) represent the highest-risk windows, and how many active customers currently sit in those windows?
- What is the revenue-at-risk if the current high-risk cohort churns at historical rates?
- Are there specific segments (plan tier, industry, company size) structurally over-represented in risk exposure?
- How concentrated is churn risk — is it a broad, even risk across the base, or concentrated in a few identifiable clusters?

4.4 Predictive Analytic — “What is likely to happen next?”

- Which active customers are most likely to churn in the next 30/60/90 days?
- What are the leading indicators (features) that best predict churn engagement decline, ticket volume spikes, payment failures, tenure stage?
- What churn probability score can be assigned to each active customer, and how does it rank-order risk relative to the composite risk score?
- How will churn rate trend over the next 1–2 quarters if current behavioral patterns continue?
- Which customer segments are predicted to have the highest future churn rates, and does this align with current diagnostic findings?

4.5 Prescriptive Analytic — “What should we do about it?”

- Given churn is experience-driven rather than price-driven, what interventions (on-boarding redesign, proactive support outreach, engagement nudges) should be prioritized over pricing/discount strategies?
- What intervention should be triggered for customers crossing into the first-six-months high-risk window?
- Which high-risk accounts should be prioritized for proactive Customer Success outreach given limited team capacity?
- What is the expected ROI/revenue-saved from intervening on the top N highest-risk accounts vs. broad-based retention campaigns?
- Should complaint-type-specific playbooks be built, and which complaint themes justify dedicated playbooks?
- What early-warning threshold (engagement score, ticket count, etc.) should trigger an automated retention workflow?

5. Hypothesis Development

Each hypothesis below operationalizes one or more business questions from Section 4 into a testable statement against the project's tables, so the Advanced SQL, AI-Assisted SQL Development, and Analytical Validation stages (stages 9–11 of the workflow) have a clear pass/fail target rather than an open-ended exploration.

H1. Price is not the primary driver of churn

H₀ (Null): Mean ARPU of churned customers equals mean ARPU of retained customers.

H₁ (Alternative): Mean ARPU of churned customers differs meaningfully from retained customers.

Fields Used: customer_churn_clean.arpu, churn_flag

Why It Matters: Directly tests the project's central re-frame (experience- vs. price-driven churn) that Objective 1 is built on; the \$34.57 vs \$34.97 gap suggests H₀ will hold, but this must be confirmed with a formal comparison rather than eyeballed means.

H2. Early tenure is a disproportionately high-risk period

H₀ (Null): Churn is evenly distributed across tenure bands.

H₁ (Alternative): A significantly higher share of churn occurs within the first 6 months of tenure than tenure's share of the active customer base would predict.

Fields Used: customer_churn_clean.tenure_at_churn, dim_customers.signup_date

Why It Matters: Underpins Objective 3; confirming this statistically (not just descriptively) justifies concentrating onboarding investment in the 0–6 month window.

H3. Engagement decline precedes churn

H₀ (Null): Engagement score trend in the 90 days before churn is no different from a matched retained cohort.

H₁ (Alternative): Engagement scores show a measurable downward trend in the 60–90 days before churn relative to retained customers.

Fields Used: fact_engagement_monthly.engagement_score, customer_churn_clean.churn_date

Why It Matters: If confirmed, engagement decline becomes a leading indicator with enough lead time to trigger a Customer Success intervention before the customer is lost.

H4. Complaint theme predicts churn risk

H₀ (Null): Churn rate is independent of complaint theme.

H₁ (Alternative): Certain complaint themes (e.g., on-boarding, integration/reliability issues) are associated with significantly higher churn rates than others.

Fields Used: fact_support_tickets.complaint_theme_id, dim_complaint_theme, customer_churn_clean.churn_flag

Why It Matters: Directly informs whether complaint-type-specific retention playbooks (Objective 4, Prescriptive Q5) are justified.

H5. Payment failures are a symptom, not a cause

H₀ (Null): Payment failures occur independently of prior engagement or ticket activity.

H₁ (Alternative): Payment failures cluster shortly before churn and are preceded by declining engagement, rather than occurring in isolation early in the relationship.

Fields Used: fact_payment_attempts, fact_engagement_monthly

Why It Matters: Resolves a specific Diagnostic question; if true, it argues against dunning/billing fixes alone as a retention lever and supports the experience-driven thesis.

H6. Risk is concentrated, not evenly spread

H₀ (Null): High behavioral risk (composite of engagement, tickets, payment failures) is evenly distributed across plan tier, industry, and company size.

H₁ (Alternative): High risk is structurally concentrated in specific segments (e.g., smaller company size, specific industries or plan tiers).

Fields Used: fact_customer_monthly_snapshot, dim_customers

Why It Matters: Determines whether Customer Success resourcing should be segment-targeted (Objective 2/4) or spread evenly.

H7. The composite risk score outperforms any single indicator

H₀ (Null): The composite behavioral risk score has no better rank-correlation with actual churn than its single strongest component (e.g., engagement score alone).

H₁ (Alternative): The composite risk score (engagement + tickets + payment failures) rank-orders actual churners more accurately than any single component in isolation.

Fields Used: fact_customer_monthly_snapshot, customer_churn_clean

Why It Matters: Justifies building and maintaining a composite score rather than a single-metric watchlist, supporting Objective 2.

H8. Stated satisfaction aligns with behavioral risk

H₀ (Null): Survey CSAT/NPS scores are uncorrelated with churn outcome.

H₁ (Alternative): Lower CSAT/NPS scores are associated with higher churn likelihood, corroborating behavioral engagement signals.

Fields Used: fact_surveys, customer_churn_clean

Why It Matters: Cross-validates behavioral data with attitudinal (voice-of-customer) data, strengthening confidence in the risk model before it drives outreach.

H9. Predicted churn probability aligns with the composite risk score

H₀ (Null): fact_churn_predictions probabilities show no meaningful rank-correlation with the composite behavioral risk score.

H₁ (Alternative): fact_churn_predictions probabilities are positively and meaningfully correlated with the composite risk score, with predictive scoring adding lift beyond the composite alone.

Fields Used: fact_churn_predictions, fact_customer_monthly_snapshot

Why It Matters: Confirms the predictive layer (Objective 2/Section 4.4) is consistent with — and adds value over — the simpler risk-scoring approach before either is operationalized.

6. Data Discovery & Data Model

6.1 Data Discovery — Domains, Grain & Business Meaning

The project's 12 tables fall into seven business domains. Each row below states the table's grain (what one record represents) and why it matters analytically.

Domain	Table(s)	Grain	Key Fields	Analytical Relevance
Customer / Account	dim_customers	1 row per customer_id	Plan tier, company size, industry, country, sign up date, churn flag/date	Anchors every fact table; defines the segments used in Objectives 1–4.
Calendar / Time	dim_date	1 row per calendar date	Date, month, quarter, year, fiscal period	Supports monthly/quarterly churn trending and tenure-window calculations.
Usage Behavior	fact_usage_monthly	1 row per customer per month	Feature usage counts, active days, login frequency	Core input for engagement decline analysis and churn prediction features.
Engagement Scoring	fact_engagement_monthly	1 row per customer per month	Composite engagement score, score trend	Direct evidence for the “experience-driven churn” thesis (Objective 1).
Customer Health Snapshot	fact_customer_monthly_snapshot	1 row per customer per month	Point-in-time MRR, tenure, status, rolled-up health indicators	Feeds the composite risk score and

Domain	Table(s)	Grain	Key Fields	Analytical Relevance
				revenue-at-risk calculations (Objective 2).
Payments	fact_payment_attempts	1 row per payment attempt	Attempt date, amount, success/fail flag, retry count	Tests whether payment failure is a cause or symptom of disengagement.
Customer Support	fact_support_tickets	1 row per support ticket	Ticket date, resolution time, complaint_theme_id, severity	Links complaint themes to churn outcomes; feeds playbook design (Objective 4).
Complaint Taxonomy	dim_complaint_theme	1 row per complaint theme	Theme name, category grouping	Standardizes free-text-style complaint reasons into an analyzable dimension.
Voice of Customer	fact_surveys	1 row per survey response	CSAT/NPS score, survey date, response text flags	Independent, attitudinal signal to validate behavioral risk indicators.
Churn Outcome (Raw)	customer_churn_raw	1 row per customer	As-loaded churn flag, churn date, source ARPU	Invalidated source-of-truth; retained for lineage and audit.
Churn Outcome (Clean)	customer_churn_clean	1 row per customer	Validated churn flag, churn date, tenure at churn, cleaned ARPU	Single trusted table for diagnostic SQL and Power BI reporting.
Predictive Output	fact_churn_predictions	1 row per active customer (scored)	Predicted churn probability, model version/date, score band	Operationalizes Objective 2/3 into a forward-looking risk ranking.

6.2 Data Model

The schema is organized as a star model with two conformed dimensions: dim_customers and dim_date shared across every fact table, so that any metric can be sliced consistently by customer attribute or by time period.

Dimension layer: dim_customers (1 row per customer) and dim_date (1 row per calendar date) provide the consistent descriptive attributes and time hierarchy used across all reporting.

Transaction fact tables: fact_support_tickets, fact_payment_attempts, and fact_surveys each record discrete events at their own grain (1 row per ticket, attempt, or survey response) and join back to dim_customers via customer_id and to dim_date via the event date.

Periodic snapshot fact tables: fact_usage_monthly, fact_engagement_monthly, and fact_customer_monthly_snapshot record one row per customer per month, enabling trend and month-over-month comparison — the structure that supports the engagement-decline and tenure-window analysis in Objectives 1 and 3.

Outcome & analytical tables: customer_churn_raw is the as-loaded source table, retained for lineage; customer_churn_clean is the validated, analysis-ready table (Section 8) that PostgreSQL diagnostic queries and the Power BI model both read from; fact_churn_predictions sits downstream of customer_churn_clean and fact_customer_monthly_snapshot, storing one scored row per active customer to support Objective 2/Section 4.4.

Supporting dimension: dim_complaint_theme normalizes free-text-style complaint categories referenced by fact_support_tickets, enabling the theme-level churn correlation used in Objective 4.

This structure lets Power BI build a single relationship model (star schema, single-direction filters from dimensions to facts) while PostgreSQL handles the heavier diagnostic joins — e.g., joining fact_engagement_monthly to customer_churn_clean on customer_id and aligning on months-before-churn — that are impractical to express purely in DAX.

7. Data Quality & Assessment

Each check below distinguishes the issue identified in the raw/source data, the action taken during cleaning (Section 8), and any limitation that remains after cleaning.

Check	Issue(s) Identified	Action Taken	Remaining Limitation
Completeness / Missing Values	Early-tenure customers (per first login) show NULL engagement scores in fact_engagement_monthly; a small number of NULL industry/company_size values in dim_customers.	NULL engagement scores imputed as 0 with an is_per_activation flag rather than dropped; NULL categorical values recoded to 'Unknown' to preserve row counts.	'Unknown' segments slightly reduce the precision of industry/size-level breakdowns.
Duplicate Records	customer_churn_raw contained a small number of repeated customer_id rows from the synthetic load process.	DE-duplicated on customer_id (latest record kept) when building customer_churn_clean; row-count reconciliation logged.	Assumes the latest record is the authoritative one; no independent source to confirm.
Data Types	Date fields (sign up_date, churn_date, ticket_date, payment_date) loaded as text; boolean-style churn/success flags loaded as 'Y'/'N' or 1/0 text.	Cast to DATE/TIMESTAMP and native BOOLEAN using CAST/CASE WHEN during the clean transformation.	None outstanding — fully standardized in customer_churn_clean and downstream fact tables.
Validity & Acceptable Ranges	A handful of negative or zero ARPU values and engagement scores outside	Values outside valid range corrected where a clear source error existed,	Flagged rows are retained in analysis with a caveat rather than excluded, to

Check	Issue(s) Identified	Action Taken	Remaining Limitation
	the expected 0–100 range.	otherwise flagged with a data_quality_flag rather than silently altered.	avoid biasing churn rate.
Categorical Consistency	Inconsistent casing/spacing in industry, country, and plan_tier (e.g., 'saas', 'SaaS', 'SaaS ').	Standardized via TRIM + INITCAP/UPPER and a controlled lookup mapping during cleaning.	New unseen category values in future loads would need the mapping extended.
Logical Consistency	A small number of records where churn_date preceded sign up_date, or ticket_date fell after churn_date.	Logically impossible records flagged with is_logically_invalid; excluded from tenure-at-churn calculations only.	Root cause (upstream synthetic generation) not fully diagnose; monitored rather than resolved.
Outliers	Extreme values in tenure (multi-year outliers in a young synthetic base) and in support ticket counts per customer.	Outliers identified via IQR bounds and flagged (is_outlier) rather than deleted, so risk-scoring logic can choose to include/exclude them.	Threshold choice is judgment-based; revisit once EDA distributions are visualized.
Referential / Relationship Integrity	A small number of orphan customer_id values in fact tables with no matching row in dim_customers (anti-join check).	Orphan rows isolated into a holding query; excluded from customer-level joins used in clean analysis.	Orphan volume is low (<1%) but not yet root-caused.
Data Limitations & Analytical Risks	Datasets is synthetic (10,000 customers); no real seasonality, macroeconomic, or competitor signal is present.	Documented explicitly as a scope boundary in this section.	Findings should be read as methodology demonstration; would need real production data to inform live business decisions.

8. Data Cleaning & Transformation (PostgreSQL / pg Admin)

customer_churn_clean was built from customer_churn_raw (and joined reference tables) through a sequence of transformation steps executed in PostgreSQL. The activities below reflect the work completed to date.

Standardization

Categorical text fields (industry, country, plan_tier, complaint_theme) were trimmed and case-normalized using TRIM combined with INITCAP or UPPER, and mapped through a controlled lookup so that variants of the same value (e.g., 'saas' / 'SaaS') collapse into one canonical label.

Missing-Value Treatment

NULLs were handled contextually rather than uniformly: engagement scores missing prior to a customer's first logged activity were set to 0 via COALESCE and flagged with is_per_activation; NULL categorical attributes were recoded to 'Unknown' using COALESCE(field, 'Unknown') so records were retained rather than dropped from segment-level analysis.

Boolean Conversion

Source flags loaded as text ('Y'/'N') or integer (1/0) — including churn_flag and payment success indicators — were converted to native PostgreSQL BOOLEAN using CASE WHEN ... THEN TRUE/FALSE, so downstream SQL could use them directly in WHERE and aggregate logic without repeated string comparisons.

Derived Fields

tenure_days and tenure_at_churn were calculated as date differences between sign up_date and either CURRENT_DATE (active customers) or churn_date (churned customers). A cleaned arpu field was derived by validating and, where necessary, recalculating revenue-per-month against payment history.

Segmentation / Bucketing

Tenure was bucketed into analysis-ready bands — 0–30, 31–60, 61–90, 91–180, and 180+ days — aligned to the 30/60/90/180-day milestones referenced in the Risk Analysis questions (Section 4.3), using CASE WHEN range logic on tenure_days.

Risk Flags

A composite high_risk_flag was constructed by combining standardized engagement score, support ticket volume, and payment failure count from fact_customer_monthly_snapshot, so that Objective 2's watch list logic could be built on a single, reusable field rather than recomputed per query.

Outlier Identification

Inter quartile-range (IQR) bounds were calculated per numeric field (ARPU, tenure, ticket count) to flag statistical outliers with is_outlier rather than removing them outright, preserving them for the Analytical Validation stage to decide inclusion/exclusion per analysis.

Post-Cleaning Validation

Row counts were reconciled between customer_churn_raw and customer_churn_clean to confirm no unintended record loss; NULL-rate audits were re-run post-cleaning on every field; and an anti-join check confirmed every customer_id in customer_churn_clean has a matching row in dim_customers.

9. Exploratory Data Analysis (EDA)

EDA has been completed against customer_churn_clean and its supporting fact and dimension tables in PostgreSQL, executed as the 8 focus areas defined in the framework below. Section 9.1 documents the full PostgreSQL query set used to generate the descriptive statistics and chart-ready outputs for each focus area; result narratives and hypothesis conclusions are recorded in Section 9.2 as the Advanced SQL and Analytical Validation stages confirm them.

Focus Area	Key Metrics / KPIs	Variables / Tables	Business Question Answered	Technique / Visualization	Expected Business Value
Churn Rate & Trend	Overall churn %, monthly/quarterly churn trend	customer_churn_clean.churn_flag × dim_date	Is churn accelerating, stable, or declining over the	Trend line / bar chart by month	Confirms or challenges the baseline 10.21% figure and

Focus Area	Key Metrics / KPIs	Variables / Tables	Business Question Answered	Technique / Visualization	Expected Business Value
			observed period?		its stability.
Segment Profiling	Churn rate by plan tier, company size, industry, country	customer_churn_clean dim_customers	Which segments over- or under-index on churn?	Stacked bar / heat map by segment	Directs where risk-scoring and playbooks should concentrate.
ARPU Distribution	ARPU distribution, churned vs retained	customer_churn_clean.arpu	Does the ARPU distribution (not just the mean) support the price-parity finding?	Box plot / histogram, churned vs retained overlay	Strengthens or nuances the “experience-driven, not price-driven” thesis.
Tenure at Churn	Distribution of tenure_at_churn; % churn within 0–6 months	customer_churn_clean.tenure_at_churn	How concentrated is churn in the early-tenure window, and where exactly does it spike?	Histogram with 30/60/90/180-day markers	Validates and sharpens the 28.1% early-churn milestone for Objective 3.
Engagement Trajectory	Engagement score trend in the 3 months pre-churn vs. matched retained customers	fact_engagement_monthly	Does engagement visibly decline before churn, and how many months in advance?	Line chart, indexed months-to-churn	Identifies the lead time available for an intervention trigger.
Support Ticket Patterns	Ticket volume and complaint theme mix, churned vs retained	fact_support_tickets dim_complaint_theme	Which complaint themes are most associated with churn?	Ranked bar chart by theme	Feeds playbook prioritization in Objective 4.
Payment Failure Timing	Payment failure rate and timing relative to churn date	fact_payment_attempts	Do failures cluster right before churn (symptom) or much earlier	Timeline / event-study style chart	Resolves the cause-vs-symptom question posed as a Diagnostic

Focus Area	Key Metrics / KPIs	Variables / Tables	Business Question Answered	Technique / Visualization	Expected Business Value
			(cause)?		question.
Survey Sentiment	CSAT/NPS distribution, churned vs retained	fact_surveys	Does stated satisfaction diverge from behavioral engagement signals?	Distribution comparison	Cross-validates behavioral risk indicators with attitudinal data.

9.1 Exploratory Data Analysis — PostgreSQL Query Set

Each focus area below carries forward its Key Metrics, Variables/Tables, Business Question, and Technique from the Section 9 framework table into the executable PostgreSQL used to produce it. Queries read from customer_churn_clean joined to the relevant dimension/fact tables, consistent with the star-schema model in Section 6.2. is_outlier and is_logically_invalid rows are excluded where a distribution or mean would otherwise be skewed, per the judgment calls logged in Section 7.

9.2 EDA Findings & Analytical Boundary

The EDA established the baseline used by the Advanced SQL stage. The strongest validated figures carried forward are a 10.21% overall churn rate, 28.40% of churn concentrated within the first six months, and near-identical ARPU for churned (\$34.57) and retained (\$34.97) customers. These findings support an experience-led investigation rather than a pricing-led explanation.

The supplied SQL contains analytical logic for the advanced questions but does not contain exported result tables for every query. Numeric results are therefore reported only where explicitly supported by the project document or SQL assumptions; unsupported segment, driver, model and forecast values are not invented.

10. Advanced SQL Analysis

The Advanced SQL stage translates the project's product questions into decision-oriented PostgreSQL analysis. The supplied script covers 32 analytical questions across descriptive, diagnostic, risk, predictive and prescriptive analytic. The emphasis is on business meaning rather than reproducing query code.

10.1 Advanced Analytical Techniques Applied

Technique	Application in the SQL	Business purpose
Window functions	LAG, moving averages, cumulative sums, RANK, ROW_NUMBER and NTILE.	Measure churn movement, rank accounts, assess risk/revenue concentration and compare life cycle observations.
CTEs	Multi-step calculations for churn, risk, prediction, complaints and thresholds.	Make complex analytical logic modular, audit-able and easier to validate.
Conditional aggregation	FILTER and CASE logic for churned/retained comparisons and event rates.	Quantify differences in customer outcomes and isolate candidate drivers.
GROUPING SETS	Multiple customer dimensions summarized in one structure.	Profile the base consistently across segment, gender, country and city.
Percentiles & dispersion	Median, standard deviation, min/max alongside averages.	Understand distributions and avoid relying on averages alone.

Lifecycle / cohort logic	Tenure bands and months-before-churn sequencing.	Locate early-life risk and deterioration windows.
Event timing	Payments, usage and engagement aligned to churn timing.	Distinguish potential leading indicators from late symptoms.
Set-based comparison	UNION-based driver comparison.	Compare price, engagement, support, satisfaction and payment signals.
JSON feature extraction	Top predictive features unpacked from model output.	Identify recurring model-important signals.
Regression projection	REGR_SLOPE and REGR_INTERCEPT on monthly churn.	Create a trend-continuation scenario for future churn.
Revenue-at-risk modeling	Probability × recurring revenue and high-risk MRR.	Prioritize retention by commercial exposure, not risk alone.
Threshold-based prescription	Behavioral bins compared with baseline churn; rule-based actions.	Turn analytical signals into repeatable early-warning workflows.

10.2 Descriptive Analytic — What Is Happening?

The descriptive layer establishes the population and baseline for every later decision. It measures overall and monthly churn, customer distribution, ARPU by segment, engagement/support/payment indicators, tenure at churn and revenue concentration.

The overall churn baseline is 10.21%. Churn is concentrated in the customer lifecycle: 28.40% of all churn occurs within the first six months. The SQL further uses a three-month moving average and month-over-month percentage-point change to distinguish underlying movement from isolated monthly variation.

Revenue concentration is assessed using ranked segment revenue and cumulative revenue share, connecting retention performance to commercial value and supporting later revenue-at-risk prioritization.

10.3 Diagnostic Analytics — Why Is It Happening?

Investigation	SQL approach	Finding / interpretation
Price vs experience	Mean, median and dispersion of ARPU by churn outcome.	Churned ARPU is \$34.57 versus \$34.97 for retained customers. The near-parity weakens a simple price-driven churn explanation.
Early-tenure churn	Tenure bands, cumulative churn share and life-cycle windows.	28.40% of churn occurs within six months, making on-boarding and early adoption a priority intervention window.
Behavioral driver comparison	Common comparison of ARPU, engagement, support tickets, CSAT and payment-failure prevalence.	Designed to identify the strongest differentiators; the supplied materials do not include final output values for every driver, so unsupported rankings are not asserted.
Engagement decline	Three-month LAG comparison and trend fields.	Tests whether deterioration precedes churn and whether there is actionable lead time.
Usage deterioration	ROW_NUMBER orders usage months relative to churn and compares logins, features used and login recency.	Tests whether disengagement becomes visible before churn.
Support / complaints	Complaint themes joined to churn outcomes and ranked by churn lift.	Identifies themes that may justify dedicated service-recovery or product playbooks.
Payment timing	Payment attempts compared with engagement and churn timing.	Tests whether failed payments are independent drivers or late-stage symptoms.
Geographic variation	Country/city aggregation of churn	Allows localization of experience

and support activity.

problems where differences are material.

10.4 Risk Analysis — Where Is the Exposure?

Risk analysis shifts the question from “why” to “where should we intervene?” The SQL uses a composite behavioral risk view built from engagement, support and payment signals, identifies high-risk active customers, calculates high-risk MRR, examines 30/60/90/180-day tenure windows and evaluates segment concentration.

NTILE-based risk decile show whether risk is concentrated in a smaller group or distributed broadly. This supports capacity-aware Customer Success operations rather than blanket outreach.

Revenue-at-risk is estimated by combining high-risk active MRR with the historical churn rate of high-risk customers. At account level, expected revenue at risk combines 90-day churn probability with ARPU, allowing accounts to be ranked by financial exposure.

10.5 Predictive Analytics — What Is Likely to Happen Next?

The supplied SQL does not train a new machine-learning model. It analyzes existing fact_churn_predictions outputs: latest 30/60/90-day probabilities, predicted risk tiers, top model features and the ranking relationship between predictive probability and composite behavioral risk.

The script also applies regression to historical monthly churn rates to project the next one and two quarters if the observed linear trend continues. This is a scenario-based trend extrapolation, not a causal forecast.

Predictive scores are most useful operationally when combined with account economics. Ranking by expected revenue at risk helps the team focus scarce retention capacity where likelihood and commercial exposure overlap.

10.6 Prescriptive Analytics — What Should We Do?

The prescriptive layer converts evidence into operational rules: first-six-month interventions, engagement/support/payment triggers, high-value account prioritization, targeted-versus-broad campaign economics, complaint-specific playbooks and early-warning thresholds.

The early-warning framework defines a threshold where observed churn is at least 1.5 times the overall baseline; ticket-based thresholds also require a minimum customer volume. It combines engagement, recent ticket activity, account value and predictive probability to select an intervention.

11. AI-Assisted SQL Development

AI-assisted SQL development supported query drafting, refinement, debugging, optimization and analytical exploration. AI helped translate product questions into measurable SQL structures, suggest CTE and window-function patterns, refine complex joins and aggregations, and surface edge cases such as NULL handling, duplicate records, division by zero and point-in-time selection.

AI also supported iterative improvement of analytical logic and readability, particularly for lifecycle comparisons, risk prioritization, revenue-at-risk calculations and threshold-based retention rules.

AI was an analytical support tool, not the source of truth. Final joins, business logic, assumptions, thresholds and interpretations were reviewed against the schema, project objectives and validation checks. The analyst retained responsibility for deciding whether the logic was appropriate and whether results were credible.

12. Analytical Validation

Analytical validation was treated as a control layer separate from query execution. A query returning rows is not, by itself, evidence that the business logic is correct.

Validation area	Validation approach	Control objective
Population reconciliation	Cross-check customer counts, churn totals and revenue against the cleaned customer table and project baseline.	Prevent unintended population changes.
Duplicate control	Use the one-row-per-customer cleaned table and check joins for row multiplication.	Prevent inflated customer, ticket or revenue metrics.
Missing / orphan checks	Use NULL audits and anti-joins established during data cleaning.	Protect customer-level joins and segment results.

Business-logic checks	Validate status, churn dates, tenure windows, latest prediction selection and month alignment.	Ensure each metric represents the intended business event.
Segment-to-total reconciliation	Compare segment counts/rates back to overall KPIs.	Confirm detailed results reconcile to headline measures.
Reasonableness checks	Review probability bounds, cumulative percentages, rankings, direction of gaps and revenue calculations.	Detect impossible or misleading outputs.
Assumption review	Separate observed values from modeled assumptions such as intervention cost and success rate.	Avoid presenting scenario estimates as actual performance.

13. Consolidated Findings

Finding	Evidence supported by the project	Business meaning
Churn is material but not primarily a pricing story	10.21% overall churn; ARPU \$34.57 churned vs \$34.97 retained.	Prioritize experience, adoption and service quality before broad discounting.
Early life-cycle is disproportionately exposed	28.40% of churn occurs within the first six months.	On-boarding and early product adoption are high-value intervention windows.
Multiple behavioral and experience signals matter	SQL explicitly tests engagement, usage, support, CSAT and payment timing.	Customer health should be monitored as a multi-signal journey.
Risk can be translated into financial exposure	SQL calculates high-risk MRR and expected revenue at risk.	Retention capacity can be allocated using commercial impact as well as likelihood.
Predictive outputs can support prioritization	Existing 30/60/90-day probabilities and risk tiers are ranked and compared.	Predictive scoring can complement behavioral health scoring after validation.
Retention can be operationalized	Thresholds, playbooks, account ranking and intervention actions are defined.	Analytics can become repeatable workflows rather than one-off reporting.

14. Product Insights

- Near-identical ARPU between churned and retained customers indicates that product experience and customer value realization deserve greater attention than price as the default churn explanation.
- The first six months should be managed as a dedicated product life-cycle stage with health checkpoints because a disproportionate share of churn occurs before durable product habits are established.
- Engagement should be monitored as a trajectory. Sustained decline can be more actionable than a single low score because it signals deterioration in value realization.
- Support activity should be treated as both a service-quality signal and a source of product-friction intelligence. Repeated or severe tickets should feed Customer Success workflows and product discovery.
- Payment failure needs contextual interpretation. Where billing friction follows declining engagement, billing remediation alone may not resolve the underlying retention problem.
- Risk prioritization should combine churn likelihood, behavioral severity and account economics so limited Customer Success capacity is directed toward the highest-impact accounts.

15. Recommendations

Priority	Recommendation	Link to evidence
Immediate	Create a first-six-month customer health journey with 30/60/90/180-day checkpoints.	Direct response to the 28.40% early-churn concentration.
Immediate	Activate alerts for low engagement, elevated recent support activity and high predictive risk.	Uses the multi-signal early-warning framework.
Immediate	Rank Customer Success outreach by expected revenue at risk.	Connects probability with ARPU and team capacity.
Short term	Build complaint-theme playbooks where volume and churn lift justify dedicated treatment.	Matches the SQL decision rule for complaint playbooks.
Short term	Strengthen on-boarding, feature-adoption walk-throughs and re-engagement journeys.	Addresses the experience-led and early-tenure churn thesis.
Short term	Feed recurring support themes into the product backlog and service-recovery process.	Turns customer friction into product improvement opportunities.
Long term	Calibrate and validate predictive risk models using live outcomes and intervention experiments.	Current SQL analyzes existing model outputs rather than training a new model.
Long term	Test pricing or discount interventions experimentally rather than using them as the default response.	ARPU parity does not support a broad price-led explanation.

15.1 Business Action Plan

Time horizon	Team	Action
0–30 days	Product + Customer Success	Launch first-six-month health monitoring and define 30/60/90/180-day intervention ownership.
0–30 days	Customer Success	Create a high-risk queue ranked by expected revenue at risk; escalate critical accounts.
0–30 days	Support + Billing	Coordinate service recovery and billing resolution for customers showing both support friction and payment failure.
30–90 days	Product	Investigate high-volume/high-lift complaint themes and convert recurring issues into product fixes.
30–90 days	Marketing + Product	Deploy lifecycle-based onboarding and re-engagement journeys.
90+ days	Data + Product	Monitor model calibration, threshold stability and intervention lift; refresh rules using production outcomes.

15.2 KPI & Impact Measurement

KPI	How to use it	Impact signal
Churn Rate	Track overall and cohort churn after interventions.	Primary retention outcome.
Early-Tenure Churn Rate	Measure churn within 30/60/90/180 days.	Tests on-boarding and early-adoption improvements.
Retention Rate	Track retained-customer share by cohort and segment.	Confirms improvement in customer survival.

Revenue Churn	Measure recurring revenue lost through churn.	Shows financial value protected, not just logos.
Net Revenue Retention (NRR)	Track retained revenue including churn, contraction and expansion.	Measures economic durability of the installed base.
MRR at Risk	Sum MRR attached to high-risk accounts.	Shows whether current exposure is shrinking.
Expected Revenue at Risk	Sum probability × recurring revenue for prioritized accounts.	Tracks forward-looking exposure.
Engagement Score / Trend	Monitor level and month-over-month change.	Leading indicator of product value realization.
Feature Adoption	Measure adoption and depth of core features.	Tests whether on-boarding creates durable usage.
Support Ticket Rate	Monitor tickets per customer and repeat-ticket behavior.	Measures service and product friction.
Complaint Churn Lift	Compare theme-specific churn with baseline.	Prioritizes playbooks and product fixes.
CSAT / NPS	Track satisfaction before/after service recovery.	Validates experience improvement.
Payment Failure Rate	Monitor failed payments and recovery.	Measures billing friction and distinguishes it from disengagement.
Intervention Save Rate	Retained after intervention ÷ customers intervened on.	Measures playbook effectiveness.
Retention ROI	Attributable revenue saved ÷ intervention cost.	Determines whether retention investment creates economic value.

16. Professional Project Summary

This project demonstrates an end-to-end Digital Product Analytics approach to SaaS customer churn: building a trusted analytical foundation, profiling customer behavior, diagnosing churn, quantifying risk and revenue exposure, evaluating existing predictive signals, and translating evidence into operational retention actions.

The central strategic implication is that the observed churn pattern is more consistent with an experience-led retention problem than a simple pricing problem. The strongest supported evidence is the 10.21% overall churn rate, 28.40% concentration of churn in the first six months, and near-identical ARPU between churned and retained customers.

Because the dataset is synthetic, these findings should be treated as a portfolio and methodology demonstration rather than production business truth. In a live SaaS environment, thresholds, predictive performance and intervention impact should be validated continuously using observed outcomes and controlled experiments.